Harvey Sykes

Fakultät für Mathematik, Universitätsstraße 150, 44801, Bochum, Germany

harvey.sykes@rub.de

Research Interests

I am a postdoc at Ruhr-Universität Bochum working under Prof. Dr. Gerhard Röhrle. My interests revolve around reductive groups and their generalisations, e.g. ind-groups and Kac-Moody groups. This usually involves a mixture of (possibly infinite dimensional) algebraic Lie theory, Coxeter theory, and building theory.

Employment

Ruhr-Universität Bochum, Wissenschaftlicher Mitarbeiter (3-year contract) — Oct 2026 – Present

Working group: Mathematik Algebra und Zahlentheorie

Lead: Gerhard Röhrle

Education

University of Essex, PhD Pure Mathematics — 2023 – 2026

Thesis title: Complete Reducibility in Kac-Moody Groups

Thesis supervisor: Alastair Litterick

Durham University, MSc Mathematical Sciences — 2022 – 2023

Grade: Distinction

Thesis title: Total Positivity

Thesis supervisor: Pavel Tumarkin

Durham University, BSc Mathematics — 2019 – 2022

Grade: Class I

Thesis title: Commutators in Rings

Thesis supervisor: Alexander Stasinski

Awards

York Mathematics Fellowship Development Bursary (£1000) — 2026

University of Essex Mathematical Sciences PhD Scholarship (3 years) — 2023

Durham University Collingwood Undergraduate Research Grant (6 weeks) — 2022

Publications

Buildings governed by twisted Bruhat orders, preprint, arXiv:2611.02925

Parabolic subgroups of ind-groups, in preparation

Invited Talks

Ruhr-Universität Bochum, Seminar Lie-Theorie — June 2026

Citadels

University of York, York Algebra Seminar — Feb 2026

Complete Reducibility in Kac-Moody Groups

University of Essex, Mathematics Essex Seminar Series — Nov 2025

Complete Reducibility in Kac-Moody Groups

University of Manchester, Heilbronn Algebra Seminar — Oct 2025
Complete Reducibility in Kac-Moody Groups

Contributed Talks

University of Essex, GGI: Latest trends in algebra and geometry — July 2026
Citadels

UCLouvain, Buildings 2025 — Sept 2025
The Dynamics of Complete Reducibility in Kac-Moody Groups

University of Bristol, Postgraduate Group Theory Conference 2025 — July 2025
Kac-Moody groups and G-complete reducibility

Warwick Venice Centre, Algebraic Groups on the Grand Canal — April 2025
The Dynamics of Complete Reducibility in Kac-Moody Groups

University of Birmingham, Postgraduate Group Theory Conference 2024 — July 2024
Kac-Moody groups over fields

Conferences Organised

University of Essex, AGGITatE 2025: Groups, Representations and Cohomology — Aug 2025
Co-organisers: Alastair Litterick, Jesus Martinez-Garcia, and Jay Taylor

Teaching

Assistant Lecturer, University of Essex — 2024 – 2026
Supporting students in weekly computer labs and marking in the following modules:

- Artificial intelligence and machine learning with applications (24/25)
- Artificial intelligence and machine learning with applications (25/26)

Outreach

Mentor, MEI — 2022
I was involved as a Mentor in the the Problem Solving Matters 2022 programme. This is a programme designed to support students from state-funded schools, academies, and colleges who are interested in studying mathematics. My responsibilities included running workshops and providing written feedback for a group of prospective university students.

Volunteer representative, PhD Your Way — 2024, 2025
I represented the University of Essex at the annual PhD Your Way event. This is an online event that provides information and advice about the PhD process to under-represented groups.

Refereeing

- Journal of Algebra and its Applications

Programming Languages

Python, R